



Recovering Hidden Parameters From Overdamped Systems

Lucas Lima and Dr. Alberto Condori
Florida Gulf Coast University – Department of Mathematics



Introduction

Damped systems arise across science and engineering, from mechanical vibrations to electrical circuits. The motion of these systems is governed by the second-order differential equation:

$$y''(t) + cy'(t) + ky(t) = 0$$

where

- c is the damping parameter,
- k is the stiffness parameter.

In practice, these parameters are *not directly measurable*. Instead, we observe the system's motion $y(t)$.

Main Question

Can we reliably recover c and k from observed data?

Parameter Estimation Method

Overdamped responses generally take the form

$$y(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$$

where r_1, r_2 are the *decay rates* of each mode and the coefficients C_1, C_2 are determined by initial conditions.

Least-Squares Formulation: Given sample data $(t_1, y_1), \dots, (t_m, y_m)$, we solve the matrix equation

$$A(r_1, r_2)x = b \quad (1)$$

in the least-squares sense, where

- $A(r_1, r_2) = [e^{r_1 t_i} \ e^{r_2 t_i}]$ is an $m \times 2$ design matrix,
- $x = (C_1, C_2)^\top$ is the vector of coefficients,
- $b = (y_1, y_2, \dots, y_m)^\top$ is the observed data.

Grid Search Strategy

To estimate the parameters c and k from observed data, we use a **grid search strategy**:

1. Construct a grid of candidate pairs (r_1, r_2)
2. For each pair, we find the least-squares solution $\hat{x}$ to (1) and compute the predicted signal $\hat{b} = A(r_1, r_2)\hat{x}$
3. Select the pair (r_1, r_2) that minimizes the error $\|b - \hat{b}\|^2$
4. Compute the parameters c and k from the selected decay rates: $c = -(r_1 + r_2)$ and $k = r_1 \cdot r_2$

Numerical Results

We plot the error in the estimated parameters over a range of possible observation windows T .

- Data is sampled uniformly over the interval $[0, T]$ with a fixed number of samples and small uniform noise ($\epsilon = 0.001$).
- The computed parameter errors are averaged over 500 different simulations.

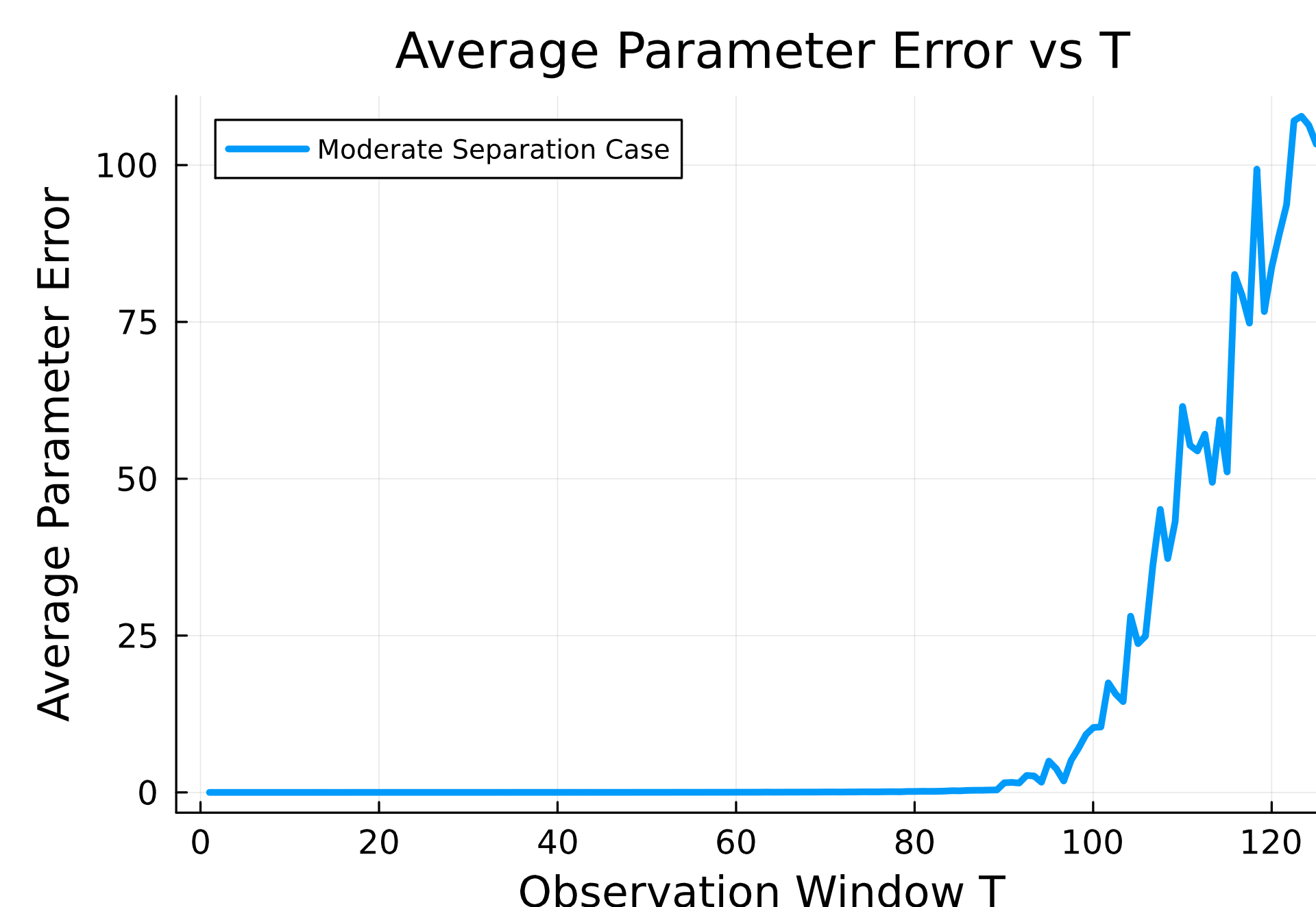


Figure 1: Data sampled from $y(t) = \frac{4}{3}e^{-t} - \frac{1}{3}e^{-4t}$ (**moderate separation in decay rates**).

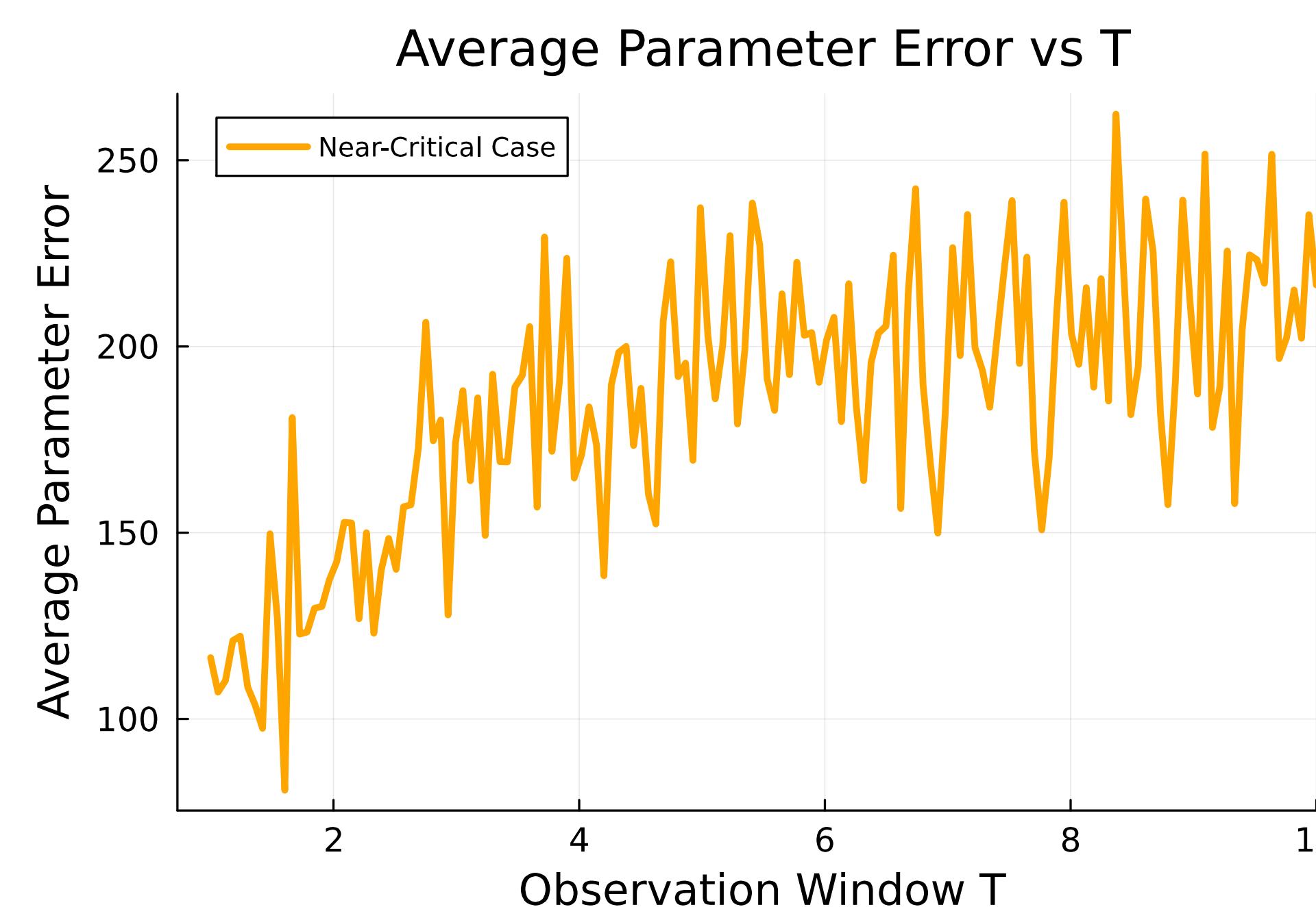


Figure 2: Data sampled from $y(t) = \frac{4}{3}e^{-1.97t} - \frac{1}{3}e^{-2.04t}$ (**nearly identical decay rates**).

Moderate separation case:

- More stable recovery
- Lower error across a wider T

Near-critical case ($r_1 \approx r_2$):

- High sensitivity
- Large estimation errors

Stable Rank

To explain these results more precisely, we introduce the **stable rank** of a matrix:

$$\text{sr}(A) = \frac{\|A\|_F^2}{\|A\|_2^2}.$$

Interpretation and role:

- Stable rank measures the **effective dimensionality** of the data matrix
- $\text{sr}(A) \approx 1 \implies$ ill-conditioned estimation

Main Theorem

Theorem 1. The stable rank of $A_n = [a_{1n} \ a_{2n}]$ approaches 1 if at least one of the following occurs:

- the column vectors a_{1n} and a_{2n} are *nearly aligned*
- one column *strongly dominates* in magnitude

Let $A_n = [e^{r_1 t_i} \ e^{r_2 t_i}]$ where $t_i \in [0, T_n]$. Then this result explains when the parameters become **unidentifiable**:

- **Near-critical case:** The columns $e^{r_1 t_i}$ and $e^{r_2 t_i}$ are nearly identical, so $\text{sr}(A) \approx 1$ for any T .
- **Moderate separation case:** One mode decays faster and data becomes more sparse as T increases. So, $\text{sr}(A) \approx 1$ for *very long* observation windows.

Note: There also exists a partial converse to Theorem 1.

Discussion & Significance

- **Estimation performance:** The grid search strategy is effective when the system is well-conditioned.
- **Identifiability insights:** The stable rank serves as a useful diagnostic for parameter identifiability.
- **Practical implications:** The effects of data sampling on identifiability highlight how careful experimental design is essential for real-world applications.

Acknowledgements

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